STAT 380 Mini Project 2 Disclosure Document

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1. Clean Data

Source: STAT 380 mini project 1

Original code started from:

```
drop_and_pivot <- function(df, value_name) {
  df |>
    select(-any_of(as.character(2023:2100))) |>
    pivot_longer(
      cols = -c(geo, name),
      names_to = "year",
      values_to = value_name
    ) |>
    mutate(year = as.integer(year))
}
sanitation_long <- drop_and_pivot(sanitation, "sanitation_access_pct")
fertility_long  <- drop_and_pivot(fertility, "children_per_woman")
co2_long        <- drop_and_pivot(co2, "co2_per_capita")
happiness_long  <- drop_and_pivot(happiness, "happiness_score")
income_long     <- drop_and_pivot(income, "income_level")
poverty_long    <- drop_and_pivot(poverty, "people_in_poverty")
population_long <- drop_and_pivot(population, "population")
```

Changes made: Used most of the original code from our previous mini project, since the variables are the same. Added democracy as a new response variable.

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2. Join Variables

Source: STAT 380 mini project 1

Original code started from:

```
#| label: join
combined <- list(
  population_long, sanitation_long, fertility_long, co2_long,
  happiness_long, income_long, poverty_long
```

```
) |>
  reduce(full_join, by = c("geo", "name", "year")) |>
  arrange(geo, year)
glimpse(combined)
#| label: check
range(combined$year)
n_distinct(combined$geo)
head(combined)
```

Changes made: Used reduce(full_join, ...) to combine the predictor datasets, filtered the result to 2020, joined the democracy data, removed incomplete rows with na.omit(), and exported the final dataset.

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3. Exploratory Data Analysis (EDA)

a. Variable 1, 4, 5

Source: https://r-graph-gallery.com/boxplot.html

Original code started from:

```
ggplot(mtcars, aes(x=as.factor(cyl), y=mpg)) + geom_boxplot(fill="slateblue", alpha=0.2) +
xlab("cyl")
```

Changes made: Replaced the example variables with Democracy and children_per_woman. Added descriptive labels and calculated the sample size, mean, median, standard deviation, minimum, quartiles, and maximum for each democracy group.

b. Variable 2

Source: AI, prompt “R code ggplot for scatterplot with crossbars”

Original code started from:

```
ggplot(combined, aes(x = Democracy, y = co2_per_capita)) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.6, color = "blue", size = 2) +
  stat_summary(fun = median, geom = "crossbar", width = 0.5,
               color = "black", linewidth = 0.4) +
  labs(title = "Democracy vs CO2 Emissions per Capita, 2020",
       x = "Democracy", y = "CO2 per capita, tonnes",
       caption = "Each dot is a country. The black bar marks the group median.") +
```

theme_minimal()

Changes made: Used Democracy as the grouping variable and co2_per_capita as the quantitative variable. Added a median marker, axis labels, and a caption. Calculated group summaries, identified extreme outliers using the 1.5 × IQR rule, and compared the two groups' median values.

c. Variable 3

Source: https://r-graph-gallery.com/48-grouped-barplot-with-ggplot2.html

Original code started from:

specie <- c(rep("sorgho" , 3) , rep("poacee" , 3) , rep("banana" , 3) , rep("triticum" , 3)) condition <- rep(c("normal" , "stress" , "Nitrogen") , 4) value <- abs(rnorm(12 , 0 , 15)) data <- data.frame(specie,condition,value) # Grouped ggplot(data, aes(fill=condition, y=value, x=specie)) + geom_bar(position="dodge", stat="identity")

Changes made: Ordered the four World Bank income categories from low to high. Created a proportional stacked bar chart and contingency tables showing country counts and proportions. Also summarized CO2 emissions by income group.

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4. kNN Models

Source: STAT 380 assignment 6

Original code started from:

insurance_recipe_train <- recipe(charges ~ sex + smoker + age + bmi + children, data = insura
step_normalize(all_numeric_predictors())|>
step_dummy(all_nominal_predictors())
insurance_recipe_train
set.seed(1855) #setting seed for reproducability
insurance_parts <- insurance |>
initial_split(prop = 0.8) #making split
insurance_train <- insurance_parts |>
training() #getting training data
insurance_test <- insurance_parts |>
testing() #getting testing data
list(insurance_train, insurance_test) |>

```
map_int(nrow) #checking sizes could do seperately and avoid list
knn_spec <- nearest_neighbor(
neighbors = tune(),
weight_func = "rectangular"
) |>
set_engine("kknn") |>
set_mode("regression")
knn_recipe <- recipe(
charges ~ age + sex + bmi + children + smoker,
data = insurance_train
) |>
step_normalize(all_numeric_predictors()) |>
step_dummy(all_nominal_predictors())
knn_wf <- workflow() |>
add_recipe(knn_recipe) |>
add_model(knn_spec)
knn_grid <- tibble(neighbors = 2:21)
tune_res <- tune_grid(
knn_wf,
resamples = val_split,
grid = knn_grid,
metrics = metric_set(rmse))
knn_final_spec <- nearest_neighbor(
neighbors = best_k$neighbors,
weight_func = "rectangular"
) |>
set_engine("kknn") |>
set_mode("regression")
knn_final_wf <- workflow() |>
add_recipe(knn_recipe) |>
add_model(knn_final_spec)
knn_model <- knn_final_wf |>
fit(data = insurance_train)
knn_train_pred <- predict(knn_model, new_data = insurance_train)
```

```
knn_test_pred <- predict(knn_model, new_data = insurance_test)
best_k
knn_model
```

Changes made: Adapted the class code to the project's democracy response and selected predictor. The model was tuned to choose k and evaluated using the project's data.

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5. Logistic Regression

Source: STAT 380 assignment 7

Original code started from:

```
spam_logistic <- glm(
spam ~
.,
data = spam_train,
family = binomial
)
spam_prob <- predict(
spam_logistic,
newdata = spam_test,
type = "response"
)
spam_pred_05 <- if_else(spam_prob >= 0.5, 1, 0)
confusion_05 <- table(
Prediction = spam_pred_05,
Truth = spam_test$spam
)
confusion_05
```

Changes made: Adapted the logistic regression model to predict Democracy using the selected predictor. Used the fitted model to examine the relationship between the predictor and the probability of democracy and evaluated classification performance.

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6. Our Question

Source: STAT 380 mini project 1

Original code started from:

```
ggplot(
  poverty_sanitation_2022,
  aes(x = people_in_poverty, y = sanitation_access_pct)
) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    x = "People in poverty",
    y = "Basic sanitation access (%)",
    title = "People in Poverty vs. Basic Sanitation Access, 2022"
  ) +
  theme_minimal()
```

Changes made: Adapted the code to the research question, selected variables, and combined dataset. The final visualization was used to support the written answer to the question.